PLAN:
For Meraz, instead of ordinary Jeopary-style CTF we have a plan to host a Cybersecurity based, real life escape room inspired CTF. Whole competition consists of let's say 10 Rooms or Levels, Each Room contains 4 challenges/puzzles each worth 25 points based on real concepts of cybersecurity, that tests your hacking skills and critical thinking, except the 10th Room which would contain a VM running image of an OS simulating an attack victim of a Worm attack, person has to gather information regarding the worm, source, etc. the more the information is valuable the more points will be granted. When a person solves all the challenges of a certain room then only he will be able to advance to the next level of room. We need to stitch each room with animated and smooth entry and exits. For Room 1 to 9 in total 4 hours will be allotted and in case anyone manages to complete all the rooms then for Room 10 1 hour more will be allotted to only those teams. Real time updates in points will occur on 0xspectre.tech dashboard. All rooms will be connected by an interesting story and nothing must feel like forcefully bolted. We need to protect infra from bruteforcing and other attacks too. Each room could have multiple different objects and NPC's to interact and examine for clues, hints etc. Creativity, Interactive, Funny must be main elements of the event.
PLANNED ARCHITECTURE:
[Players] --ssh--> [Bastion] --> [LANDING CONTAINER]-
[R1]-[R2]-[R3]-[R4]-[R5]-[R6]-[R7]-[R8]-[R9]-[R10:VM] --solves--> [Scoring API] --ws--> [Website Scoreboard]

Here, each room is a docker container which contains:

1. Object, Challenge, NPC etc rendering engine of that room
2. Art of that room
3. Verification Logic of that room

ESTIMATED TOOLS:
Level-1:

1. OpenSSH
2. Fail2ban : for bruteforce detection
3. Iptables
4. Tmux : proposed plan is that whole team will share same terminal, if possible we can make it terminals per teammate synced in real time (will be complex)
5. Asciinema : records terminal activity to intercept cheating

Level-2:

1. Docker : each room is a docker container
2. Docker Compose : each team will get fresh bag of docker containers
3. Cgroups : to manage core and resources of containers
4. Firecracker or equivalent : to enhance security (Optional)

Level-3:

1. blessed/urwid/rich python library and pyfiglet : to add beauty, objects, characters etc to room.
2. Chafa : for ASCII art to add to the terminal

Level-4: For final Room 10

1. QEMU
2. Libvirt
3. Packer

UNPOLISHED DEMO of a Room: Art is not up to the mark nor challenge.

IMAGE-1: SORT OF LANDING PAGE

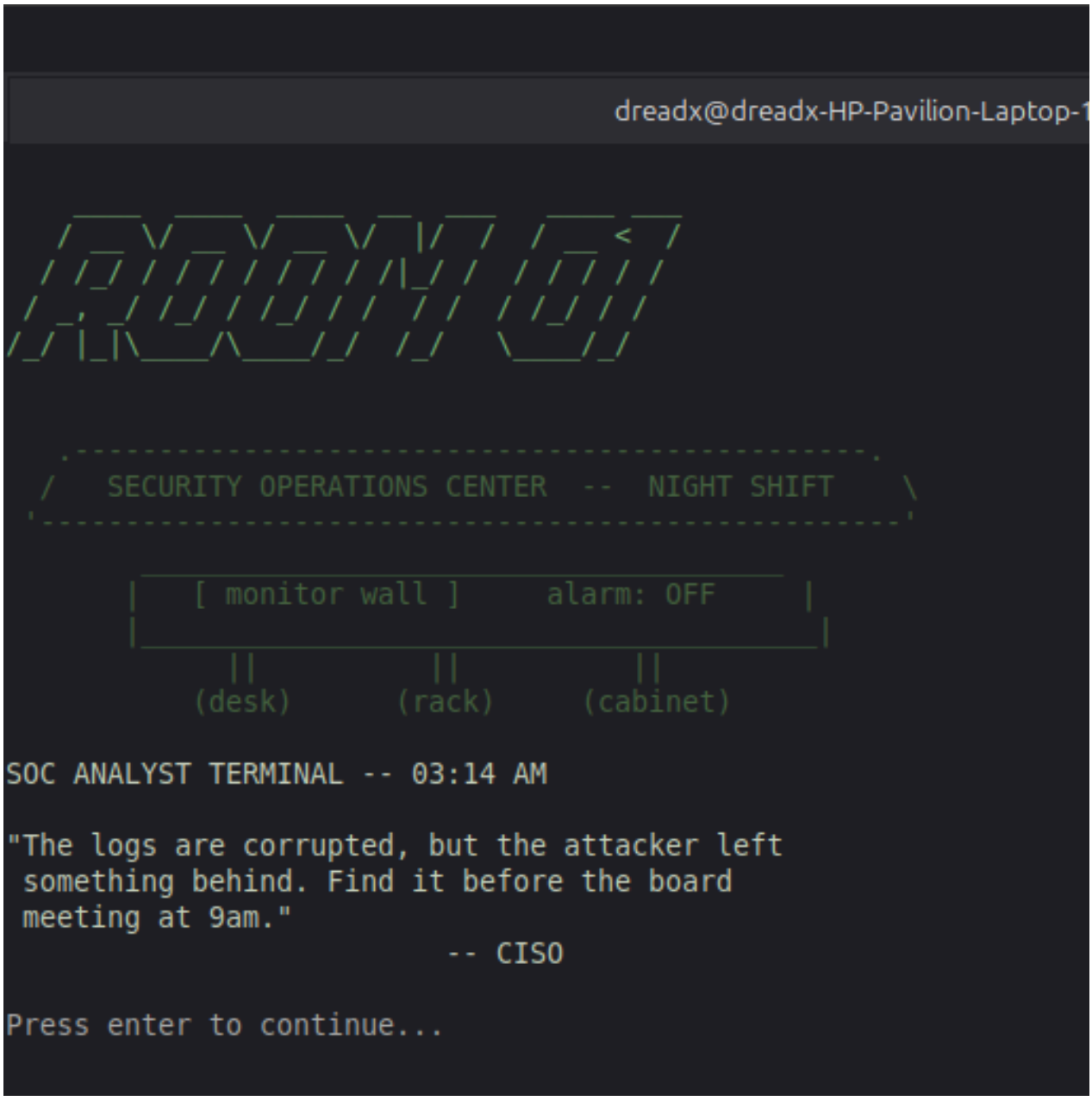
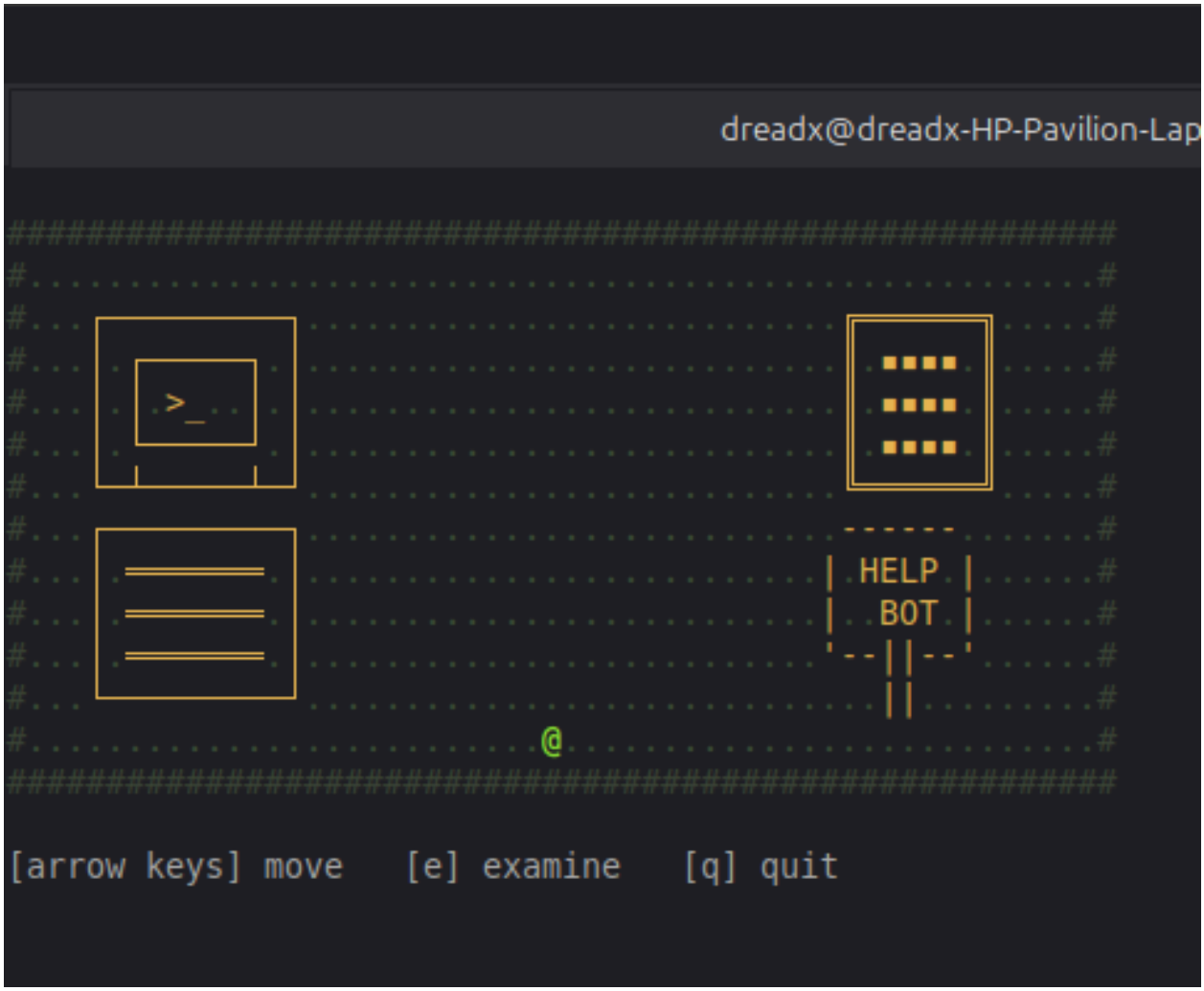
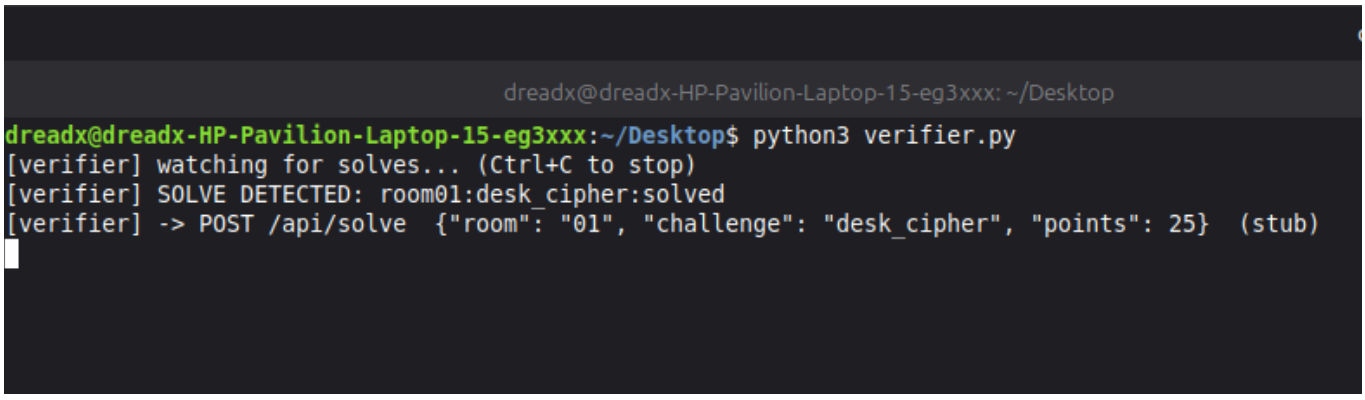


IMAGE-2: ROOM - ‘@’ is players character can interact with objects when goes near them.





ESTIMATED TIME: 6 weeks
SOURCE CODE (100% AI generated) WILL BE ATTACHED WITH THIS DOC !!
HOW TO RUN DEMO:
$ pip install blessed pyfiglet
$ mkdir art
And send .txt files into this art/

$ python3 engine.py
In second terminal
$ pyhton3 verifier.py